

第四周作业参考解答

p.76, 习题 12.1, 1(2), (3), 2, 4, 5(1), 6, 7, 8.

1. 解: (2) 令 $F = (z^2 - 2a^2)x^2 + a^2y^2$, 则 $\frac{\partial F}{\partial x} = 2x(z^2 - 2a^2)$, $\frac{\partial F}{\partial y} = 2a^2y$, $\frac{\partial F}{\partial z} = 2zx^2$,

因此 $\left. \frac{\partial z}{\partial x} \right|_{(a,a,a)} = -\frac{\frac{\partial F}{\partial x}(a,a,a)}{\frac{\partial F}{\partial z}(a,a,a)} = 1$, $\left. \frac{\partial z}{\partial y} \right|_{(a,a,a)} = -\frac{\frac{\partial F}{\partial y}(a,a,a)}{\frac{\partial F}{\partial z}(a,a,a)} = -1$, 故曲面在 (a, a, a) 处的

法线方程为 $\frac{x-a}{1} = \frac{y-a}{-1} = \frac{z-a}{-1}$, 切平面方程为 $x - y - z + a = 0$.

(3) 令 $x_0 = u_0 \cos v_0$, $y_0 = u_0 \sin v_0$, $z_0 = av_0$. 设曲面在 (x_0, y_0, z_0) 处的法向量为

(A, B, C) , 其中

$$A = \begin{vmatrix} \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \\ \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} \end{vmatrix}_{(u_0, v_0)} = a \sin v_0, \quad B = \begin{vmatrix} \frac{\partial z}{\partial u} & \frac{\partial z}{\partial v} \\ \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \end{vmatrix}_{(u_0, v_0)} = -a \cos v_0, \quad C = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix}_{(u_0, v_0)} = u_0,$$

则曲面在 (x_0, y_0, z_0) 处的法线方程为: $\frac{x-x_0}{a \sin v_0} = \frac{y-y_0}{-a \cos v_0} = \frac{z-z_0}{u_0}$;

切平面方程为: $a \sin v_0 (x - x_0) - a \cos v_0 (y - y_0) + u_0 (z - z_0) = 0$.

2. 解: 曲面的法方向为: $\left(\frac{\partial F}{\partial x}, \frac{\partial F}{\partial y}, \frac{\partial F}{\partial z} \right) = \left(\frac{2x}{a^2}, \frac{2y}{b^2}, \frac{2z}{c^2} \right)$, 由题意可知 $P(x_0, y_0, z_0)$ 处的

法向量满足: $\frac{2x_0}{a^2} = \frac{2y_0}{b^2} = \frac{2z_0}{c^2}$, 令 $x_0 = \lambda a^2$, $y_0 = \lambda b^2$, $z_0 = \lambda c^2$, 则由点 $P(x_0, y_0, z_0)$

在曲面上, 带入曲面方程解得 $\lambda = \pm \frac{1}{\sqrt{a^2 + b^2 + c^2}}$, 故 P 点坐标为

$$\left(\frac{a^2}{\sqrt{a^2 + b^2 + c^2}}, \frac{b^2}{\sqrt{a^2 + b^2 + c^2}}, \frac{c^2}{\sqrt{a^2 + b^2 + c^2}} \right)$$

$$\text{或} \left(-\frac{a^2}{\sqrt{a^2 + b^2 + c^2}}, -\frac{b^2}{\sqrt{a^2 + b^2 + c^2}}, -\frac{c^2}{\sqrt{a^2 + b^2 + c^2}} \right).$$

4. 解: 曲面 $z = x^2 + y^2$ 在切点 $P(x_0, y_0, z_0)$ 处的法向量为 $(2x_0, 2y_0, -1)$.

直线 l 显然与两平面的法向量 $(1, 0, 2)$ 和 $(0, 1, 2)$ 正交。因此直线 l 的方向向量为:

$$\vec{v} = \begin{vmatrix} i & j & k \\ 1 & 0 & 2 \\ 0 & 1 & 2 \end{vmatrix} = (-2, -2, 1),$$

由切平面与直线垂直知曲面的法向量 $(2x_0, 2y_0, -1)$ 与直线 l 的方向向量 $\vec{v} = (-2, -2, 1)$ 平行,

所以 $x_0 = y_0 = 1, z_0 = 2$, 故切平面为 $2x + 2y - z = 2$.

5. (1) 证明: 曲面 $z = yf\left(\frac{x}{y}\right)$ 在 (x_0, y_0, z_0) (其中 $z_0 = y_0 f\left(\frac{x_0}{y_0}\right)$) 处的法向量为

$$\left(\frac{\partial z}{\partial x}, \frac{\partial z}{\partial y}, -1\right)\bigg|_{(x_0, y_0, z_0)}, \text{ 其中 } \frac{\partial z}{\partial x}(x_0, y_0, z_0) = f'\left(\frac{x_0}{y_0}\right),$$

$$\frac{\partial z}{\partial y}(x_0, y_0, z_0) = f\left(\frac{x_0}{y_0}\right) - \frac{x_0}{y_0} f'\left(\frac{x_0}{y_0}\right),$$

因此切平面方程为:

$$f'\left(\frac{x_0}{y_0}\right)(x - x_0) + \left[f\left(\frac{x_0}{y_0}\right) - \frac{x_0}{y_0} f'\left(\frac{x_0}{y_0}\right)\right](y - y_0) - (z - y_0 f\left(\frac{x_0}{y_0}\right)) = 0,$$

即 $f'\left(\frac{x_0}{y_0}\right)x + \left[f\left(\frac{x_0}{y_0}\right) - \frac{x_0}{y_0} f'\left(\frac{x_0}{y_0}\right)\right]y - z = 0$, 故切平面恒过坐标原点. 证毕

6. 解: 由于空间曲线 l 是两个曲面的交线, 因此曲线在一点处的切线为两曲面在该点处切平面的交线。两曲面在 $P(1, -2, 1)$ 处的法向量分别为: $\vec{n}_1 = (2, -4, 2)$, $\vec{n}_2 = (1, 1, 1)$, 所以曲线 l

在 $P(1, -2, 1)$ 处的切线方向向量 $\vec{v} = \vec{n}_1 \times \vec{n}_2 = (-6, 0, 6)$, 故曲线 l 在 $P(1, -2, 1)$ 处的切线方

程为 $\begin{cases} x + z = 2 \\ y = -2, \end{cases}$ 法平面方程为 $z = x$.

7. 证明: 螺旋线的切线方向向量为 $\vec{v} = (x', y', z') = (-a \sin t, a \cos t, b)$, 设其与 z 轴正方向

$(0, 0, 1)$ 的夹角为 θ , 则有 $\cos \theta = \frac{b}{\sqrt{a^2 + b^2}}$ 为定值, 故切线与 z 轴夹角为定值. 证毕

8. 证明: 若直线 l 为曲面 S 上一条曲线在 P 点处的切线, 则其为两个曲面在 P 点处切平面的交线。记直线 l 的方向向量为 $\vec{v}$, 又曲面 S 在 P 点的法向量为 $\vec{n}_1 = (f'_x, f'_y, f'_z)\big|_P$, 则另一曲

面在 P 点处切平面的法向量为 $\vec{n}_2 = \vec{v} \times \vec{n}_1$. 任取一过 P 点且在 P 点法向量为 $\vec{n}_2$ 的曲 (平)

面 Q (或直接取平面 $\vec{n}_2 \cdot (x - x_0, y - y_0, z - z_0) = 0$), 可知其与曲面 S 的交线 L 在 P 点处的切线为 l . 证毕

p.82, 习题 12.2, 1(1), (2), 2(1), 3.

$$1. \text{ 解: } (1) \frac{\partial f}{\partial x}(0,0)=1, \frac{\partial f}{\partial y}(0,0)=1, \frac{\partial^2 f}{\partial x^2}(0,0)=\frac{\partial^2 f}{\partial y^2}(0,0)=\frac{\partial^2 f}{\partial x \partial y}(0,0)=0,$$

所以, 当 $x^2 + y^2 \rightarrow 0$ 时, $f(x, y)$ 在 $(0, 0)$ 点可展开为带有皮亚诺余项的二阶的泰勒公式

$$f(x, y) = x + y + o(x^2 + y^2).$$

$$\text{由于 } \frac{\partial^2 f}{\partial x^2}(x, y) = \frac{\partial^2 f}{\partial y^2}(x, y) = \frac{\partial^2 f}{\partial x \partial y}(x, y) = -\sin(x + y), \text{ 因此对任意的 } (x, y) \in \mathbb{R}^2,$$

$f(x, y) = \sin(x + y)$ 在 $(0, 0)$ 点可展开为带有拉格朗日余项的一阶的泰勒公式

$$\sin(x + y) = x + y - \frac{1}{2} \sin(\theta x + \theta y)(x + y)^2, \quad \theta \in (0, 1).$$

解法二、令 $t = x + y$, 则 $z = f(x, y) = \sin t$, 也可以用一元函数的 Taylor 公式得到相同的结果.

(2) 由于 $z = e^{x^2 - y^2} \in C^\infty$, 因此其任意阶混合偏导数与求偏导的顺序无关, 直接计算得

$$\frac{\partial z}{\partial x} = 2xe^{x^2 - y^2}, \quad \frac{\partial z}{\partial y} = -2ye^{x^2 - y^2},$$

$$\frac{\partial^2 z}{\partial x^2} = (4x^2 + 2)e^{x^2 - y^2}, \quad \frac{\partial^2 z}{\partial y^2} = (4y^2 - 2)e^{x^2 - y^2}, \quad \frac{\partial^2 z}{\partial y \partial x} = -4xye^{x^2 - y^2},$$

故, 当 $x^2 + y^2 \rightarrow 0$ 时, $z = e^{x^2 - y^2}$ 在 $(0, 0)$ 点可展开为带有皮亚诺余项的二阶的泰勒公式

$$z = 1 + x^2 - y^2 + o(x^2 + y^2).$$

对任意的 $(x, y) \in \mathbb{R}^2$, $z = e^{x^2 - y^2}$ 在 $(0, 0)$ 点可展开为带有拉格朗日余项的一阶的泰勒公式

$$e^{x^2 - y^2} = 1 + [2\theta^2(x^2 - y^2)^2 + x^2 - y^2]e^{\theta^2(x^2 - y^2)}, \quad \theta \in (0, 1).$$

2. (1) 解: 直接计算, 有

$$\frac{\partial z}{\partial x}(1, 1) = yx^{y-1} \Big|_{(1,1)} = 1, \quad \frac{\partial z}{\partial y}(1, 1) = x^y \ln x \Big|_{(1,1)} = 0,$$

$$\frac{\partial^2 z}{\partial y^2}(1,1) = x^y (\ln x)^2 \Big|_{(1,1)} = 0, \quad \frac{\partial^2 z}{\partial x^2}(1,1) = y(y-1)x^{y-2} \Big|_{(1,1)} = 0,$$

$$\frac{\partial^2 z}{\partial x \partial y}(1,1) = (1+y \ln x)x^{y-1} \Big|_{(1,1)} = 1,$$

$$\frac{\partial^3 z}{\partial x^3}(1,1) = y(y-1)(y-2)x^{y-3} \Big|_{(1,1)} = 0, \quad \frac{\partial^3 z}{\partial y^3}(1,1) = x^y (\ln x)^3 \Big|_{(1,1)} = 0,$$

$$\frac{\partial^3 z}{\partial x^2 \partial y}(1,1) = [2y-1+y(y-1)\ln x]x^{y-2} \Big|_{(1,1)} = 1,$$

$$\frac{\partial^3 z}{\partial x \partial y^2}(1,1) = (2 \ln x + y \ln^2 x)x^{y-1} \Big|_{(1,1)} = 0,$$

故 $z = x^y$ 在 $(1,1)$ 点的三阶 Taylor 多项式为 $T_3(x, y) = x + (x-1)(y-1) + \frac{1}{2}(x-1)^2(y-1)$,

$$\text{且 } (1.1)^{1.02} \approx 1.1 + 0.1 \times 0.02 + \frac{1}{2} 0.1^2 \times 0.02 = 1.1021.$$

3. 证明: 设 $f(x, y) = \frac{\cos x}{\cos y}$, 则 $f(x, y)$ 在原点附近无穷次可微, 且

$$f(0,0)=1, \quad f'_x(0,0)=0, \quad f'_y(0,0)=0, \quad f''_{xx}(0,0)=-1, \quad f''_{yy}(0,0)=1, \quad f''_{xy}(0,0)=0.$$

当 $|x|$ 和 $|y|$ 充分小时, $\frac{\cos x}{\cos y}$ 可由其二阶泰勒多项式近似, 即 $\frac{\cos x}{\cos y} \approx 1 - \frac{1}{2}x^2 + \frac{1}{2}y^2$. 证毕